

reduced by 1 value.

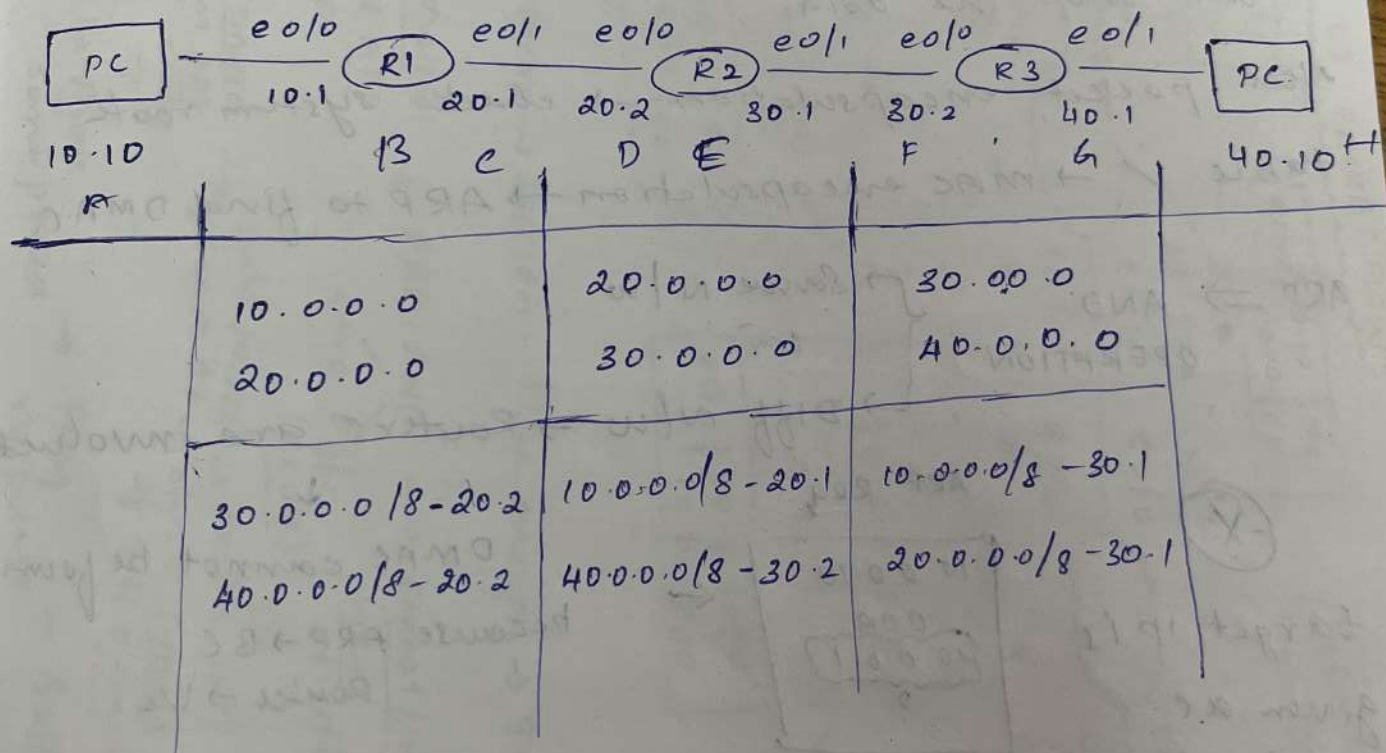
If the TTL comes a value

If the TTL comes with zero $\rightarrow$ stop looping
data will drop.

If the TTL becomes zero then will get
a error message: Time expired.

TTL expired.

Day-13 Routing in depth. (X) Ping 40.0.0.10



If we ping 40.0.0.10 echo $\rightarrow$ Format & time encaps. $\rightarrow$
packet encapsulation $\rightarrow$ check system route table $\rightarrow$ MAC encaps.
 $\rightarrow$ Dest. MAC? $\rightarrow$ ARP REQ $\rightarrow$ But ARP is broadcast, Router
is unicast, so we cannot find destination MAC
directly. In ARP Req. put default gateway as
target IP.

When ARP comes then AND operation is done to find whether the devices are connected in same or different network. If different network then it understands router is present. So when sending ARP req. default gateway is given as target IP.

→ After giving ARP reply from router it notes the target MAC in ARP table & submits to ethernet protocol.

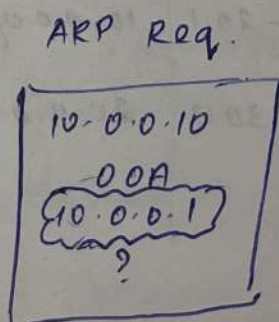
→ Router will first check whether the data's destination MAC is its own, if it is the router's dest. MAC it will open or else it will drop the data.

After packet encapsulation → check system route table ✓ → MAC encapsulation → ARP to find D-MAC.

ARP ⇒ AND OPERATION

- Same N/w (or)
- Diff. N/w → Routers are involved.

target IP is given as default gateway



DMAC cannot be found because ARP → BC
Router → VC

→ Route (✓)
But DMAC we don't know

→ ARP Req / Rep (x8)

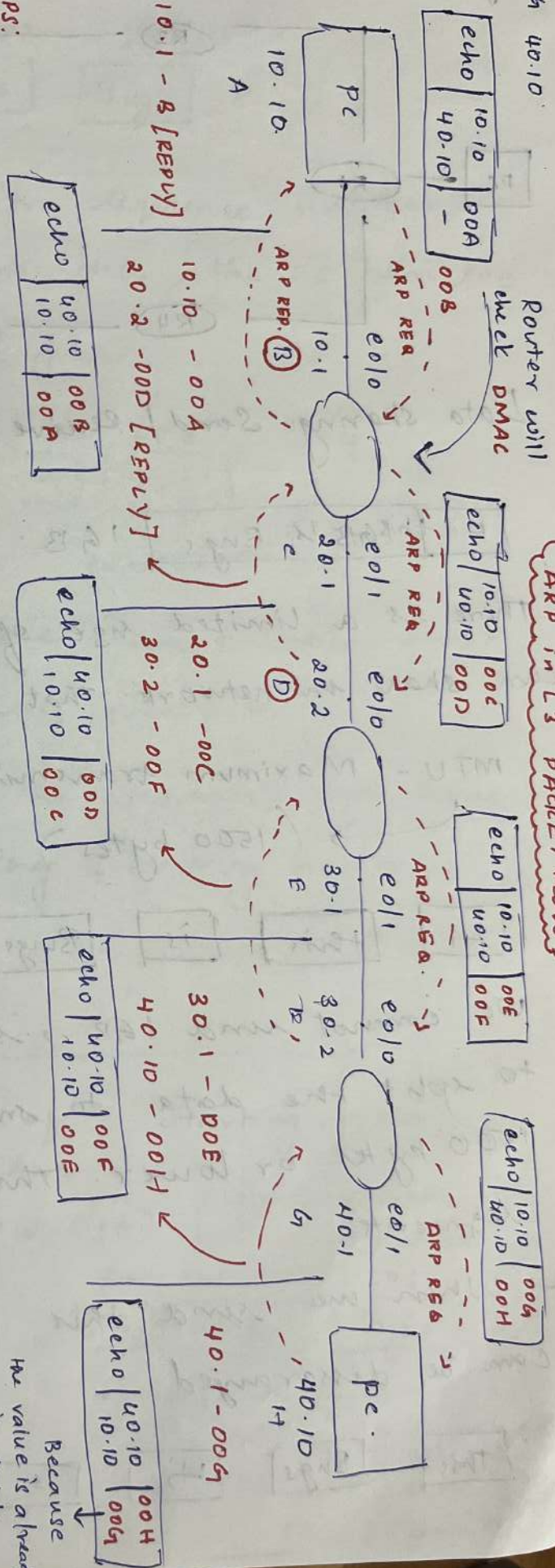
→ 40.10

same problem

PIN 4 40.10

Router will check DMAC

ARP in L3 PACKET FLOW



STEPS:

Ping 40.10 → App-layer-Echo → Pres.-Format encap → Session-Time encap → Trnsp. → N/w-Packet

→ check destination → MAC encap → will call ARP for help
IP in system route
table (✓)
SIP ✓
DIP ?

→ Req. will go to R1 → R1 will note & give reply
Note DMAC from ARP reply
Data packet → Router will check DMAC (check DMAC decapsulation route)

→ Route (✓) → ARP Req (Rep (x)) → It reaches 40.10 → Again same process → Ping response → But no ARP req. because we know from ARP table.

TIP: 206

Because the value is already in table